The Open Source Firefighter

Coordinated Defense Tactics — A Combined Arms Paper

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Andrew Morton · Palmer Lake, Colorado

This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

This paper describes how coordinated ember defense fights — alone at each tier, and combined with the fire service's existing arms. It is doctrine, not requirement: the specification says what the equipment does; this paper says how the fight uses it. It mirrors the equipment family deliberately: three tiers, one architecture, and the combined operation at the end. It is written for fire officers, planners, and the people who will stand behind the line.

> **A treatise on a new interface firefighting technology that enables the incident commander with a strategic resource at the line of contact.**

The premise

Wildland firefighting has always been combined arms. Hand crews cut line, engines hold it, dozers build it faster, aviation cools what the ground cannot reach, and command orchestrates the pieces against a fire that gets a vote. Coordinated ember defense does not replace any arm of that fight. It adds one: **ground that defends itself** — water, timed and placed by machines that never get tired, commanded by people who do. The tactical question this paper answers is what that new arm does at each scale, and what the old arms are freed to do because of it.

One idea runs through every tier: **the receiving surface is the battle.** Structures are lost to embers landing on dry fuel; ground is lost to fire crossing dry fuel. Keep the receiving surface wet during the window when it matters, and the fire arrives at a place that will not have it. Everything below is that one idea at three different sizes.

Tier One — The Structure (the box on the wall)

The unit of action: one node, one structure, one to three zones. The residential wall unit or its commercial sibling, wetting roofs, gutters, wall lines, fences, and the vegetation ring on a rotating cycle driven by live drying conditions.

What it fights: ember cast — the attack that destroys most structures in an interface fire. Firebrands landing on dry receiving surfaces, ahead of and behind the front, in quantities no crew can chase. A wet roof sheds the attack that a dry roof accepts.

How it fights: sequencing. One zone at a time, so an ordinary household supply protects a whole structure; re-wet intervals computed from the fuel's actual drying rate under actual weather, so water is spent exactly as fast as the physics demands and no faster. The node fights alone if it must — no network, no command, no grid.

Tactical effect: a defended structure is subtracted from the list of places a person must stand. That is the whole effect, and it is the most important sentence in this paper. Every structure that holds itself releases the crew that would have held it.

Bounding: a node defends against ember ignition and surface-fuel attack. It does not defeat direct flame impingement or radiant load from a crowning slope at close range — a structure inside that geometry is defended at Tier Three or evacuated, not promised a miracle.

Tier Two — The Community (the mesh and the module)

The unit of action: the coordination domain — a street, a neighborhood, a town — plus the Coordinated Defense Module: one Type 6 engine, a trailer of prepositioned cases, a small element, a drone pilot; ordered and typed like any other resource.

What it fights: the mathematics of simultaneous demand. Every structure needs water in the same hour; the shared

supply cannot feed them all at once; uncoordinated defense collapses the main and protects no one. And it fights the coverage gap — the unprotected parcel, the unarmed house, the vacant lot that turns one ignition into a block.

How it fights: the domain schedules itself — structures take turns on the shared supply automatically, holding total draw within what the mains deliver, yielding instantly when an engine company needs the water (a pressure sag is an order every node obeys without being told). The module extends and repairs the domain: it arrives reading the town's own census off the mesh, seeds nodes into the mapped gaps at open-lid tempo, adds detection overwatch and command presence, and moves on — leapfrogging cluster to cluster, because the nodes hold ground after the module leaves. Prepared communities multiply the module; the load plan writes itself from the domain census in the first minute of radio contact.

Tactical effect: the staffing arithmetic inverts from crews-per-site to sites-per-crew. A town that would have consumed twenty engines in static structure protection is held by its own boxes plus a handful of engines working exceptions. The exceptions are where people belong: the woodpile spot, the pre-code roof, the un-boxed unit.

Bounding: participation density is the controlling variable at suburban spacing — below roughly thirty feet of separation, structures ignite each other, and one protected home in five is a wet spot where three in five is a firebreak. The community tier is only as strong as its adoption map, and the gaps concentrate exactly where money is thinnest. Doctrine must say so, because the fire will.

Tier Three — The Terrain (the wet line)

The unit of action: the pre-surveyed defensive network — nodes emplaced on the ground the fire must cross: the drainages, chokepoints, saddles, and field edges that a century of fire history names in advance. Sited in the quiet season, written into the community's protection plan, cached against the corridors they will hold.

What it fights: the approach itself. Not the fire at the walls — the fire three drainages out, funneling toward town on the same roads it has always used.

How it fights: the reversible fireline. A continuous wetted strip is a constructed line that builds itself faster than hand or dozer line and is held by nodes instead of crews. Its tactical function is identical to fuel-removal line — deny the fire the next foot of receiving fuel — achieved by temporarily raising fuel moisture instead of permanently destroying vegetation. Wetting a fine fuel below ignition threshold is forestry-manual arithmetic; the water needed to hold a narrow line against surface fire is small, and it comes from what the ground offers — a millpond, a stock tank, static storage, a tender loop. The network is **passive until triggered**: emplaced at negligible standing cost, dormant most days of most years, deploying alongside the resident mesh only when conditions and a front arrive. Presence is separated from staffing, which is what makes landscape-scale line economically possible where continuous hose-lay staffing never was.

Green islands are the tier's signature move: a pre-wetted patch ahead of the front is a break point where the fire arrives with less energy — and once committed and safe, it is a helicopter landing zone. The same water builds the line and the airport the incident will need.

Tactical effect: the fight moves off the structures entirely. A front broken or steered at the corridor never becomes a thousand simultaneous ember problems in town. And when it is over, the line evaporates: the vegetation lives, the ground is unscarred, the community was not destroyed to be saved — which dissolves the political resistance that limits destructive precautionary line.

Bounding (mandatory, always stated): the wet line holds surface fire and flanking/backing fire — which is most of a fire's perimeter, most of the time, and where most acres are actually held. It does **not** hold a wind-driven crown run spreading by radiant heat and long-range spotting; no crew stakes its egress on a wet line; the line is best-effort infrastructure that buys probability, not certainty; and employment is human-commanded at every step. The system is a tool a qualified officer wields. It does not fight fire on its own.

The Combined Operation

No tier fights alone in a real event, and the tiers were designed to compose. What follows is the shape of the combined fight — the new arm integrated with the old ones.

Days out — shaping the battlefield. The terrain tier deploys first, because ground does not get wet in an afternoon. Corridors go active off the pre-survey; static sources are verified (*verified* — a walk-around is not an inspection, and the fire will find the February shortcut with terrible precision); the tender loop stands up; green islands saturate and become tomorrow's LZs. Residents arm and self-test; the mesh reports the town's readiness as one picture; modules arrive against the CWPP priority and seed the mapped gaps. Aviation gets its fill points and

landing zones from the same survey. **Water logistics is the campaign** — the boxes are the easy part; the tender driver on the ninth climb of a fourteen-hour shift is the hard part, and the plan must budget roads, fills, and fatigue like the consumables they are.

The smart water grid, sparse by design. The manual water-handling layer remains the backbone; remotely commanded appliances (the hydrant valve actuator, smart gates and wyes, the multi-port manifold) are placed only at selected key junctions where command flexibility pays — chosen by the IC, advised by hydraulics and the system's own ledger history. The signature employment is the **buffered supply point**: portable tanks, a smart manifold in a case, and an observation node watching levels turn a tender shuttle into managed standing storage — a water tower assembled from shuttle loads, with refills and tank swaps triggered on observed actuals. Supply state becomes visible on the command surface, a withdrawing crew can close the hydrant behind itself from the truck, and the supply officer commands the grid instead of driving to its valves. Every smart appliance works manually when dead; the smart version is a powered hand, never a replacement for one.

The approach — fighting at the corridor. The wet line meets the front at the tier-three ground: holds it where the water reached, turns it along the wetted edges, stalls it against the islands. Crews do not stand at this line — they patrol it, tend it, and chase what spots over it, with the drone walking the perimeter when the air allows and the fixed detection carrying the watch when it doesn't. Engines and dozers work where doctrine still needs them — anchor points, contingency line, the places wet line is the wrong tool. Command watches one picture: every node, every tank level, every corridor, live. The forward weather nodes buy the minutes that matter when the outflow comes — and the outflow will come. And command has a second picture available on demand: a thermal pass over the domain reads protection state directly off the ground — cool roofs holding, hot roofs not — so the incident commander sees which structures are actually wet and which have failed or gone unprotected, at range, without pulling a crew off the line to check. It confirms the coordination picture against physical reality: where the map says wet and the roof reads hot, that is where the next problem is, and command knows it before the fire does.

The arrival — the shrunken fight in town. What crosses the corridors arrives at the community tier pre-shrunk: ember cast and leftovers instead of a front. The domain runs its rotation; the mains hold because the schedule holds; municipal-supplied nodes yield the instant an engine drafts. Structure crews — a fraction of what static protection would have consumed — spread out and monitor the whole defense, working the exceptions: the gaps in the adoption map, the pre-code construction, the ember bed the rotation reached one beat late. This is combined arms at its purest: the boxes are the line, the engines are the maneuver element, and each is free to be excellent at its half because the other half is covered.

After — the honest accounting. The ledger holds the whole event: every gallon, every cycle, every yield, every failure logged the hour it happened. The after-action writes itself from a record that cannot be shaded, the wet line disappears into ordinary ground, the cases go home signed against their cards, and the findings — the split tank, the chewed shoulder, the coin-flip coverage map — go back into February, where the next fire is already being decided.

Command integration, stated once: the system presents one command surface at every tier — a residential node and a terrain-line node in the same picture, distinguished by capability, never by separate applications — and publishes into the mapping tools commanders already run. Above it, any battle-management layer the era provides (common operating pictures, drone and sensor fusion) can task and read the network through the same closed vocabulary, because the box is control-system agnostic by construction. And the security is self-sorting by prior design: a node cannot execute a command for hardware it does not carry — the capability bits define each node's executable subset — so the attack surface of any node is exactly its installed equipment and nothing more. Operators prevail over automation at every tier, always, per the charter. The machines hold the water. The people hold the judgment.

The experienced-eyes doctrine. The common operating picture makes seasoned judgment deployable at the cost of a glance, and the service is full of judgment currently trapped behind desks and knees. Three postures, one mechanism: **the desk assist** — the FMO or fire staff officer who came up on the jumps and now runs a small department cannot peel off for every little start, but with the picture already rendering beside the budget spreadsheets, assisting collapses from a half-day drive to a two-minute look and one radio call ("watch that draw — it spots across every time; set your trigger point now"), delivered to every initial attack in the district at zero marginal staffing, in exactly the window — small fire, thin resources, green IC deciding alone — where one experienced set of eyes has always been worth the most. **The defensive seat** — the coordinated defense renders as a screen-runnable domain (nodes, corridors, tank levels, coverage, the wet line: allocation and timing, not chainsaws), and doctrine assigns it as a defensive group under a seasoned officer, including the retired or limited-mobility chief who knows the county better than anyone but can no longer run the woods — freeing the incident commander to concentrate on the main attack while span of control improves for both. **The consultation** — the old hand in front of the live picture, opinion on request; the profession that consumed his knees keeps his judgment. In every posture the mechanism is the same and the boundary is the same: the system does not create the expertise or the responsibility — it deletes the travel between them. Augmentation, never command transfer; on-scene authority retained, per the charter, always.

The one-page version, for the officer in a hurry

- **Tier One:** the box holds the structure. Every structure that holds itself releases a crew.
- **Tier Two:** the mesh holds the town's water math; the module fills the gaps and moves on. Crews-per-site becomes sites-per-crew. Participation density is the controlling variable.
- **Tier Three:** the wet line holds the ground the fire must cross — reversible, pre-surveyed, passive until triggered, human-commanded, honest about crown fire.
- **Combined:** shape days out, fight at the corridor, mop the shrunken fight in town, and read the whole thing off one ledger afterward. Water logistics is the campaign. February is where the fire is actually fought.
- **The point:** not machines replacing firefighters — reach and leverage handed to the people who already do the work, so fewer of the places on fire require a person standing there.

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Appendix A — Supply & Coverage Planning Table

Formerly OSF-2.2, folded into this paper at consolidation; printable as a standalone extract. The register records the supersession.

Conservative planning figures only. Everything is site-specific: wind, slope, elevation gain, and obstructions shrink every number below. The unit's commissioning self-calibration measures your actual supply and governs; this page is for planning, not performance guarantees.

Table 1 — What your supply run honestly delivers

Assumes 50 psi municipal/well pressure at the spigot, level ground. Figures rounded down on purpose.

Supply run	50 ft	100 ft	200 ft
1/2" garden hose	~3 GPM @ ~30 psi	~2–3 GPM @ ~22 psi	~2 GPM @ ~15 psi
5/8" garden hose	~5 GPM @ ~40 psi	~4 GPM @ ~32 psi	~3 GPM @ ~22 psi
3/4" garden hose	~6 GPM @ ~44 psi	~5–6 GPM @ ~38 psi	~4–5 GPM @ ~30 psi
3/4" poly/PEX/pipe	~7 GPM @ ~45 psi	~6 GPM @ ~40 psi	~5 GPM @ ~34 psi
1" pipe	~8+ GPM @ ~47 psi	~8 GPM @ ~44 psi	~7 GPM @ ~40 psi

Rule that teaches itself: over distance, hose diameter is everything. Borrowing supply across the lot line? **Borrow the fat hose.** Subtract ~5 psi per 10 ft of elevation gain.

Table 2 — What each sprinkler class needs and covers

Derated field figures, not catalog still-air numbers. Effective wetted radius assumes light wind.

Head class	Needs (min)	Appetite	Effective radius	Use for
Micro/low-flow spray or mini-impact (plastic)	~15 psi	0.5–2 GPM	8–15 ft	Gutter line, deck, fence line, spot targets
Half-pattern spray head	~20 psi	1–2 GPM	10–12 ft	Wall line, roof edge from below
Small plastic impact	~25 psi	2–3 GPM	15–25 ft	Small roofs, yard corners, thin supply runs
Brass impact, forestry standard (25PJ class)	~40 psi	3–5 GPM	25–35 ft	Roof ridge, broad coverage — the structure-protection workhorse

Table 3 — Spacing: how far apart to put them

The table above gives each head's reach. This one says where the next head goes. Wildland guidance says to place heads so adjacent patterns overlap; irrigation practice states the same rule as a number.

Spacing rule	Distance between heads	When to use
Head-to-head (the standard)	equal to the effective radius	Baseline. Each head throws to the base of the next; patterns overlap by half.
Wind-derated	~90% of the effective radius	Any exposed site, any forecast wind — which is most fire days.
Hard wind	~80% of the effective radius	Ridge tops, canyon mouths, sustained wind.

Do not space at the full diameter of throw. Two heads set a full diameter apart leave a dry hole between them, and the dry hole is where embers sit. Spacing equal to *radius*, not diameter, is the whole rule.

Design pressure is 50 psi. Cache-standard structure-protection heads are built around it. Below roughly 40 psi the radius collapses and every spacing figure above becomes wrong — the layout that overlapped on paper no longer overlaps on the ground. Check Table 1 for what your run actually delivers before trusting a spacing.

Trunk and laterals, not one long hose. Avoid feeding several heads from a single straight run of 3/4 in. hose; the last head starves. Run the largest line you have as a trunk and branch laterals off it. The rule scales down intact: on a single-property install, this means running the fat line to a manifold near the structure and short laterals to each head.

Table 4 — Match line

One zone runs at a time. Plan heads per zone, not per system.

Your supply (from Table 1)	Workable head classes	Heads per zone
~2 GPM @ low psi (long thin run)	Micro/low-flow only	1–2
~3–4 GPM @ ~20–30 psi	Micro, half-pattern, small plastic impact	1–2
~5–6 GPM @ ~35–45 psi	Any above; one brass impact	1 brass or 2–3 small
~7+ GPM @ ~40+ psi	Full forestry pattern	2 brass impacts

Rules box

1. **Receiving sites beat area.** Wet the gutters, roof valleys, fence line, and the junk pile — not the lawn. Embers land on things.
2. **Rotation is the design.** One zone at a time is how a garden spigot protects a whole structure. Do not plan for simultaneous zones.
3. **The forestry anchor:** the brass impact class above is the same sprinkler wildland crews deploy for structure protection. If it reaches your ridge line at your pressure, use it there.
4. **The thin-supply anchor:** if Table 1 puts you in the top rows, small plastic impacts and micro-heads placed *at the targets* out-protect one starving brass head every time.
5. **The undefended parcel next door is coverable:** 200 ft of 3/4" hose and two well-chosen small heads will keep it wet. Not well — enough.
6. **Site specifics govern.** Wind, slope, trees, and your actual measured pressure override every figure on this page. The unit measures truth at commissioning.

Sources: supply figures derive from published hose and pipe friction-loss tables and manufacturer minimum operating specifications, derated for field conditions. Head classes, placement practice, and the overlap rule adopt published wildland structure-protection sprinkler guidance — the national cache kit standards and the structure-protection field literature — by reference to their current editions. Spacing and precipitation-rate relations follow standard irrigation design practice (Irrigation Association / manufacturer design guidance). This page is a planning aid and is not a substitute for the specification.